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**3 Complex Analysis 2023**

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**Test 1**

30 August 2023

**Instructions:**

- This question paper consists of **5 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 40 points = 100% (42 points available).

| Problem | 1 | 2 | 3 | 4 | $\Sigma$ | Mark |
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| Maximum Points | 12 | 10 | 10 | 10 | 40 | 100% |
| Result | | | | | | |

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**Problem 1**

(12 points)

(a) Let $G \subseteq \mathbb{C}$ be an open set and $f : G \to \mathbb{C}$. Let $z \in \mathbb{C}$. Define what is meant by $f$ being differentiable at $z$, and define the derivative of $f$ at $z$.

(b) Show by definition that the function $f(z) = |z|^2$ is differentiable at $z = 0$, and determine $f'(0)$. (Hint: the definition of the limit can be helpful in considering the limit here.)

(c) Show by a method of your choice that $f(z) = |z|^2$ is not differentiable at any $z \neq 0$.

**Solution.**

(a) $f$ is differentiable at $z$ if the limit $\lim_{h\to 0} \frac{f(z+h)-f(z)}{h}$ exists. In this case, the derivative of $f$ at $z$ is defined by

$$f'(z) = \lim_{h\to 0} \frac{f(z+h) - f(z)}{h}.$$

(b) We have

$$\lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{|h|^2}{h}.$$

We claim that $\lim_{h\to 0} \frac{|h|^2}{h} = 0$. Indeed, given $\varepsilon > 0$,

$$0 < |h| < \varepsilon \implies \left| \frac{|h|^2}{h} - 0 \right| = |h| < \varepsilon.$$

Thus, $f$ is differentiable at $z = 0$, and $f'(0) = 0$.

(c) We are going to use the Cauchy-Riemann equations $u_x = v_y$, $u_y = -v_x$ that give a necessary condition for differentiability. We have $f(z) = |z|^2 = x^2 + y^2$, so that $u(x, y) = x^2 + y^2$ and $v(x, y) = 0$. The derivatives are

$$u_x = 2x, \quad u_y = 2y, \quad v_x = 0, \quad v_y = 0.$$

The Cauchy-Riemann equations take the form

$$2x = 0, \quad 2y = 0.$$

The only solution is $(x, y) = (0, 0)$. It follows that $f$ is not differentiable at any $z \neq 0$.

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**Problem 2**

(10 points)

(a) Determine the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n} x^{2n}$.

(b) Find the power series expansion of the form $\sum_{n=0}^{\infty} a_n z^n$ for the function $f(z) = \frac{z}{2+z^2}$, and determine its domain of convergence. (Hint: you may use without proof known facts about known series.)

**Solution.**

(a) The coefficients are

$$a_m = \begin{cases} \frac{(-1)^n}{2n}, & n = 2m \text{ with } m \in \mathbb{N}, \\ 0, & n = 2m-1 \text{ with } m \in \mathbb{N}. \end{cases}$$

For the radius of convergence we have

$$\frac{1}{R} = \limsup_{m\to\infty} |a_m|^{\frac{1}{m}} = \lim_{n\to\infty} \left(\frac{1}{2n}\right)^{\frac{1}{2n}} = \lim_{n\to\infty} e^{\frac{1}{2n}\ln\left(\frac{1}{2n}\right)} = e^0 = 1.$$

Thus, $R = 1$.

(b) We have

$$f(z) = \frac{z}{2+z^2} = \frac{z}{2}\frac{1}{1-\left(-\frac{z^2}{2}\right)} = \frac{z}{2}\sum_{n=0}^{\infty}\left(-\frac{z^2}{2}\right)^n = \sum_{n=1}^{\infty}(-1)^n\frac{z^{2n+1}}{2^{n+1}}.$$

The geometric series converges if and only if $\left|-\frac{z^2}{2}\right| < 1$, which is equivalent to $|z| < \sqrt{2}$. The domain of convergence is $\{z \in \mathbb{C} : |z| < \sqrt{2}\}$.

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**Problem 3**

(10 points)

Recall that $\cos z = \frac{e^{iz}+e^{-iz}}{2}$, $\sin z = \frac{e^{iz}-e^{-iz}}{2i}$, $z \in \mathbb{C}$.

(a) Prove that $\cos^2 z + \sin^2 z = 1$.

(b) Show that the function $\sin z$ is unbounded on $\mathbb{C}$.

(c) Recall that $\cos z = \sum\limits_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}$. Show that this series converges absolutely and uniformly on any closed disc $\{z \in \mathbb{C} : |z| \leq r\}$, where $r > 0$.

**Solution.**

(a) We have

$$\begin{aligned}\cos^2 z + \sin^2 z &= \left(\frac{e^{iz}+e^{-iz}}{2}\right)^2 + \left(\frac{e^{iz}-e^{-iz}}{2i}\right)^2 \\ &= \frac{1}{4}(e^{i2z} + 2e^{iz-iz} + e^{-i2z}) - \frac{1}{4}(e^{i2z} - 2e^{iz-iz} + e^{-i2z}) \\ &= \frac{1}{4} \cdot 4e^0 = 1.\end{aligned}$$

(b) Take $z = it$ with $t \in \mathbb{R}$. Then $\sin(it) = \frac{1}{2i}(e^{-t} - e^{t})$, so that $|\sin(it)| = \frac{1}{2}|e^t - e^{-t}|$. The later term is unbounded on $\mathbb{R}$.

(c) First we consider the series $\sum_{n=0}^{\infty} \frac{r^{2n}}{(2n)!}$, where $r > 0$. We apply the Ratio Test. We have

$$L = \lim_{n\to\infty} \frac{r^{2(n+1)}}{(2(n+1))!}\frac{(2n)!}{r^{2n}} = \lim_{n\to\infty} \frac{r^2}{(2n+2)(2n+1)} = 0.$$

Therefore, the series $\sum_{n=0}^{\infty} \frac{r^{2n}}{(2n)!}$ converges.

Now assume that $|z| \leq r$. Then $\left|(-1)^n \frac{z^{2n}}{(2n)!}\right| \leq \frac{r^{2n}}{(2n)!}$ for all $n = 0, 1, 2, \ldots$. By above and by the Weierstrass M-Test, $\sum\limits_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}$ converges absolutely and uniformly on the closed disc $\{z \in \mathbb{C} : |z| \leq r\}$.

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**Problem 4**

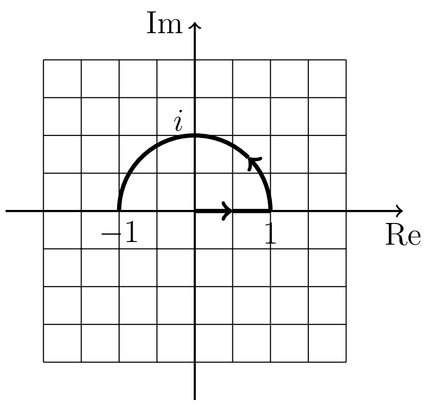
(10 points)

Consider the path

$$\gamma(t) = \begin{cases} t, & t \in [0,1], \\ e^{i\pi(t-1)}, & t \in [1,2]. \end{cases}$$

(a) Sketch the path in the graph paper below. Don't forget to label the axes and to show the orientation of the path.

(b) Calculate $\int_\gamma z^2\, dz$.




**Solution.**

(a) See the graph above.

(b) We have

$$\begin{aligned} \int_\gamma z^2\, dz &= \int_0^1 t^2\, dt + \int_1^2 e^{i2\pi(t-1)} i\pi e^{i\pi(t-1)}\, dt = \frac{1}{3} + i\pi \int_1^2 e^{i3\pi(t-1)}\, dt \\ &= \frac{1}{3} + i\pi \int_0^1 e^{i3\pi u}\, du = \frac{1}{3} + i\pi \int_0^1 (\cos 3\pi u + i \sin 3\pi u)\, du \\ &= \frac{1}{3} + i\pi \left( \frac{\sin 3\pi u}{3\pi}\Big|_0^1 - i\, \frac{\cos 3\pi u}{3\pi}\Big|_0^1 \right) \\ &= \frac{1}{3} + \frac{\pi}{3\pi}(\cos 3\pi - \cos 0) = \frac{1}{3} + \frac{1}{3}(-1-1) = -\frac{1}{3}. \end{aligned}$$